

Decoder ladder for sparse codebooks

From heuristic global unrolling to calibrated, candidate-local geometric inference

Stage	6 of 6
Updated	10 September 2026
Status	D0–D4 definitions and test contract; comparative performance evidence pending

This report is a chronological research record. Measured results, interpretations, and open hypotheses are labelled separately. Detailed numerical ledgers remain in **results/**.

1 Scope and common model

After matched-filter combination across receive antennas, all five learned decoders use

$$y = Ca^* + z, \quad C = [c_1, \dots, c_M] \in \mathbb{C}^{n \times M}, \quad a^* \in \mathbb{Z}_+^M, \quad \mathbf{1}^T a^* = K.$$

Here K is known to every current decoder. The intended constrained codebooks have unit-energy columns but need not be orthogonal. D0–D2 retain the full $M = 2^B$ message axis. D3 and D4 instead receive a bounded candidate list and generate only its columns. The cost of producing that list is outside those decoders and must be reported separately.

Conditioned on K uniform transmissions, the joint count vector is multinomial and one marginal is

$$A_m \sim \text{Binomial}(K, 1/M).$$

For $K/M \ll 1$, collisions are negligible and this reduces to a known-cardinality Bernoulli support model. The Poisson law with rate K/M is another discrete approximation to the same marginal, but it allows counts above K and discards the fixed-total dependence. D2–D4 therefore use Bernoulli inference; the local section decoder in Report 4 retains its exact Binomial count denoiser. A Poisson denoiser remains appropriate when the population size itself is random.

The decoder progression has two branches:

$$\begin{aligned} \text{D0} &\xrightarrow{\text{global learned context}} \text{D1}, \\ \text{D0} &\xrightarrow{\text{analytic effective channel}} \text{D2} \xrightarrow{\text{bounded candidates}} \text{D3} \xrightarrow{\text{learned geometry}} \text{D4}. \end{aligned}$$

D1 asks how much an expressive full-message correction can improve performance. D2–D4 instead separate the analytic, scalable, and expressive changes so that their effects can be measured.

2 Common data-consistency step

Let a^t be the soft estimate at layer t . Every decoder preserves the physical forward and adjoint operations

$$r^t = y - Ca^t, \quad g^t = \Re\{C^H r^t\}.$$

Thus gradients pass through the actual transmitted codewords. D0 and D1 form the learned gradient step $u^t = a^t + \eta_t g^t$; D2–D4 use a calibrated sufficient statistic derived below. Hard nonnegative integer projection is used only for evaluation.

3 D0: learned scalar calibration

D0 treats every coordinate as the same scalar channel. It forms one variance proxy per received sample by blending residual energy with the known physical noise variance, then uses

$$\ell_j^t = \text{clip}_{[-30,30]} \left(\gamma_t \frac{u_j^t - 1/2}{\tau_t} + \beta_t \log \frac{\rho}{1 - \rho} + b_t \right), \quad \rho = K/M.$$

The proposal is $p_j^t = \sigma(\ell_j^t)$, rescaled so that its total mass is K , then damped with the previous state. Each layer learns step size, residual/noise mixture, variance scale, evidence gain, prior gain, bias, and damping.

D0 is cheap and supplies the baseline shared by all later decoders. Its limitations are also precise: one scalar variance cannot represent codeword-dependent interference; prior gain, evidence gain and temperature are not uniquely identified; and multiplying all probabilities to obtain mass K is not a Bernoulli posterior projection.

4 D1: expressive global correction

D1 keeps D0 unchanged up to its base logit. A feature vector for message j contains u_j^t, a_j^t, g_j^t , the base logit, residual/noise statistics, and factor occupancy. A learned analysis map produces a latent h_j^t . Fixed-number attention slots pool latent context within messages sharing a factor pattern, within messages sharing a local value, and across the whole message set. A learned correction is then added:

$$\ell_j^{t,D1} = \ell_j^{t,D0} + \Delta_\theta(h_j^t, \text{pooled context}).$$

The final correction head is initialised to zero, so an untrained D1 equals D0 exactly. Its fixed-slot implementation is linear rather than quadratic in M , but it still stores and processes all M message states. D1 is therefore an expressive small- B benchmark, not the scalable receiver. Its correction can improve ranking, but its logits need not remain calibrated likelihood ratios.

5 D2: analytic effective-channel calibration

D2 returns to D0 and replaces its scalar variance heuristic. It starts from the prior mean $a_j^0 = \rho$, rather than D0's zero state, so the initial residual error is mean zero under the Bernoulli approximation. Define the real decision matrix

$$\bar{C} = C \quad \text{for real } C, \quad \bar{C} = \begin{bmatrix} \Re C \\ \Im C \end{bmatrix} \quad \text{for complex } C, \quad G = \bar{C}^T \bar{C} = \Re\{C^H C\}.$$

With $d_j = G_{jj}$, one adjoint coordinate is exactly

$$g_j^t = d_j(a_j^* - a_j^t) + \sum_{k \neq j} G_{jk}(a_k^* - a_k^t) + \bar{c}_j^T \bar{z}.$$

Adding back the known self-term gives the decision statistic

$$\zeta_j^t = g_j^t + d_j a_j^t = d_j a_j^* + \xi_j^t, \quad \xi_j^t = \sum_{k \neq j} G_{jk}(a_k^* - a_k^t) + \bar{c}_j^T \bar{z}.$$

Its inactive and active means are 0 and d_j . Under a diagonal, conditionally uncorrelated error approximation, its variance is

$$\omega_{j,t}^{\text{ana}} = \nu d_j + \sum_{k \neq j} G_{jk}^2 v_k^t, \quad \nu = \begin{cases} \sigma^2, & \text{real noise,} \\ \sigma^2/2, & \text{circular complex noise.} \end{cases}$$

Here $v_k^t = a_k^t(1 - a_k^t)$. Equal variances alone do not make this exact: fixed- K projection correlates coordinates, and reusing the same observation across layers introduces dependencies unless an extrinsic or Onsager correction is available.

The implementation avoids both the full $n \times M$ codebook and the $M \times M$ Gram matrix. Selected columns are visited in bounded chunks to accumulate a small real frame. Let

$$s_j = \sum_k G_{jk}^2 = \bar{c}_j^T (\bar{C} \bar{C}^T) \bar{c}_j, \quad \bar{v}_{-j}^t = \frac{1}{M-1} \sum_{k \neq j} v_k^t.$$

The implemented mean-field approximation is

$$\hat{\omega}_{j,t} = \alpha_t \left[\nu d_j + \bar{v}_{-j}^t (s_j - d_j^2) \right], \quad \alpha_t > 0,$$

where α_t absorbs model mismatch. Under the equal-variance Gaussian channel, the likelihood ratio is

$$e_{j,t} = \log \frac{p(\zeta_j^t | A_j = 1)}{p(\zeta_j^t | A_j = 0)} = \frac{d_j(\zeta_j^t - d_j/2)}{\hat{\omega}_{j,t}}.$$

D2 can equivalently be derived from $u_j^t = a_j^t + \eta g_j^t$: both the separation of its two means and its variance scale by η and η^2 , so η cancels exactly. D2 therefore has no learned step size and requires no spectral norm estimate. Its only learned quantities per layer are α_t and damping.

D2 adds the known prior only after computing this evidence. Rather than multiplying probabilities by a common factor, it finds the unique scalar λ_t satisfying

$$p_j^t = \sigma \left(e_{j,t} + \log \frac{\rho}{1-\rho} + \lambda_t \right), \quad \sum_j p_j^t = K.$$

This is a product-Bernoulli projection onto the known expected cardinality, not the exact joint fixed- K posterior. It keeps every p_j^t in $[0, 1]$. D2–D4 do not clip their evidence logits. For a uniform prior, the constant $\log(\rho/(1-\rho))$ is absorbed by λ_t ; its substantive roles are to initialise a^0 and to make the extension to nonuniform extrinsic priors explicit. D2 remains a global M -state reference. Its analytic geometry is cached for a frozen encoder; joint encoder learning recomputes the two chunked $O(n^2 M)$ frame passes. Its current construction is not a $B = 100$ decoder.

6 D3: candidate-restricted D2

Let a proposer return $K < Q \ll 2^B$ payloads

$$\mathcal{C}_b = \{w_{b1}, \dots, w_{bQ}\}, \quad C_b = [\phi(w_{b1}), \dots, \phi(w_{bQ})] \in \mathbb{C}^{n \times Q}.$$

D3 applies the complete D2 recurrence with C_b and a Q -length state. Its current implementation forms the bounded $Q \times Q$ candidate Gram once per forward pass; this is independent of 2^B but must still be counted. The compact hash/amplitude generator from Report 5 constructs each selected column on demand. D3 itself therefore creates neither an $n \times 2^B$ codebook nor a 2^B state. The complete path is M -free only with the direct procedural generator; the small- B comparison adapter still stores global row and label lookups. Without a calibrated extrinsic proposer prior, D3 uses the neutral conditional prior K/Q . Proposal scores derived from the same observation cannot simply be added as independent evidence without double counting.

The restricted physical model is

$$y = C_b a_b^* + \epsilon_b, \quad \epsilon_b = z + \sum_{w \notin \mathcal{C}_b} a_w^* \phi(w).$$

If every active payload is proposed, $\epsilon_b = z$ and D3 isolates the effect of replacing the global axis by a bounded candidate axis. If an active payload is absent, no D3 update can recover it; known-cardinality projection can then force mass onto a false candidate. Every non-oracle evaluation must therefore report proposal recall and Q before PUPE.

Exhaustive small- B candidates certify equivalence with D2. Oracle-completed lists give a labelled upper bound on candidate-conditioned recovery. Neither is evidence of a scalable proposer. D3 becomes a complete scalable receiver only when paired with a measured, sub-exhaustive, high-recall proposal algorithm.

7 D4: learned candidate-geometry correction

D4 adds expressivity only after D3 has removed the global message axis. Its node features contain the D3 state, decision statistic, channel evidence, effective variance, column energy, candidate density and optional proposal score. The gradient is omitted because $g_j^t = \zeta_j^t - d_j a_j^t$ is redundant. Every candidate pair has the normalised Gram coefficient

$$\tilde{G}_{ij} = \frac{c_i^H c_j}{\|c_i\|_2 \|c_j\|_2}.$$

For the generated hash family, edge features may additionally contain the fraction of common hash bins and whether the two messages share an amplitude-bank label. With node embedding h_i^t and edge feature e_{ij} , the implemented permutation-equivariant correction has the form

$$\begin{aligned} \gamma_{ij}^t &= \tanh f_\theta(e_{ij}), & m_i^t &= \frac{\sum_{j \neq i} \gamma_{ij}^t V_\theta h_j^t}{\max\{1, \sum_{j \neq i} |\gamma_{ij}^t|\}}, \\ \tilde{\Delta}_{i,t} &= o_\theta(h_i^t + g_\theta(h_i^t, m_i^t, \bar{h}^t)), & \Delta_{i,t} &= \tilde{\Delta}_{i,t} - \frac{1}{Q} \sum_{j=1}^Q \tilde{\Delta}_{j,t}, & e_{i,t}^{\text{D4}} &= e_{i,t}^{\text{D3}} + \Delta_{i,t}, \end{aligned}$$

where $\bar{h}^t$ is global candidate context. The output head o_θ is bias-free and zero-initialised, so D4 initially equals D3 exactly. Centering removes the candidate-common logit direction that known- K projection cannot identify; without it, BCE could reward a shift that changes neither probabilities nor decisions. Cardinality projection and damping then proceed as in D2.

D4 can learn effects missed by the mean-field variance, including signs and interactions of particular correlations. It can also overfit to candidate-list composition, destroy evidence calibration, or spend $O(Q^2)$ work learning an effect already captured analytically. D4 is therefore the performance-oriented candidate benchmark; D3 is its causal control.

8 Training objective

D0 and D1 historically use class-balanced support BCE with a Smooth-L1 count auxiliary. D2 and D3 train their analytic pre-prior channel evidence; D4 trains the corrected pre-prior discriminant, which need not remain a calibrated LLR:

$$\mathcal{L}_{\text{ev}} = \frac{1}{2} \frac{1}{|S|} \sum_{j \in S} \text{BCE}(e_j, 1) + \frac{1}{2} \frac{1}{|S^c|} \sum_{j \notin S} \text{BCE}(e_j, 0),$$

with weighted deep supervision over layers. This prevents the learned variance from cancelling the rare-event prior. Let $K_C = \sum_j a_j^*$ be the target mass on the represented axis. The count auxiliary is

$$\mathcal{L}_{\text{count}} = \frac{1}{\max(K_C, 1)} \sum_j \text{SmoothL1}(a_j, a_j^*), \quad \mathcal{L} = \mathcal{L}_{\text{ev}} + 0.1 \mathcal{L}_{\text{count}}.$$

For global or recall-one candidate targets, $K_C = K$. If the list misses a transmission, D3/D4 still project to mass K while the target contains less mass; this deliberate failure diagnostic is not a suitable training target. Current candidate-learning checks therefore use oracle-complete lists. There is no activity ℓ_1 penalty. Under the known- K constraint, $a_j \geq 0$ and $\mathbf{1}^T a = K$ already imply $\|a\|_1 = K$, so such a penalty would be constant.

9 Complexity and experimental meaning

Let $\mathcal{A}(P)$ denote one forward/adjoint pair for P retained columns, d a latent width, S a fixed attention slot count, and T the number of unrolled layers.

Decoder	State and cost	Learned part	Purpose
D0	M ; $O(T[\mathcal{A}(M) + M])$	Layer scalars	Scalar PGD baseline
D1	M ; D0 + $O[T(Md^2 + MSd)]$	Global correction	Value of full-axis context
D2	M ; D0-like; fixed setup $O(n^2M)$	Scale and damping	Analytic calibration
D3	Q ; $O(nQ^2 + T[\mathcal{A}(Q) + Q])$	D2 scalars	Bounded refinement
D4	Q ; D3 + $O[T(Q^2d + Qd^2)]$	Graph correction	Learned residual geometry

Comparisons should pair encoder, channel samples, layer count and stopping rule. D2 should have D0-like layer cost after geometry setup, fewer parameters and cleaner calibration, but D1 has much greater small- B capacity. D3 should equal D2 with exhaustive candidates and approach it as recall tends to one. Any trained D4 gain must justify its runtime and calibration cost relative to its exactly equal initial D3 state.

D4 ablations should change one mechanism at a time: exact $\sum_{k \neq j} G_{jk}^2 v_k^t$ variance; full covariance or Onsager/extrinsic correction; top- k Gram edges; Gram-only versus hash/label features; local LMMSE preconditioning; or shared recurrent weights.

Evidence status: D0/D1 are established small- B baselines; D2–D4 are an incremental test ladder. Local checks certify implementation, not improved performance. Candidate-conditioned decoding is not scalable complete-message recovery until its proposer is non-oracle and measured.